

Customer Churn Analysis

Task 4 – Segmentation and Recommendation Model

Objective: Move from descriptive analytics to an explainable prioritization model using customer account and support features.

1. Executive Summary

The dataset contains **15 customers**. 7 customers are marked as churned and 8 as retained, giving a churn rate of **46.67%**. Average tenure is **18.80 months**. The rule-based model assigns each customer a 0–100 risk score and places customers into Low, Medium, or High priority segments. Monthly charges associated with currently churned customers total **■409.93** in the supplied sample.

2. Features Used

The model uses contract type, tenure in months, support-ticket volume, subscription tier, and the observed churn flag. Customer ID and gender are retained for identification/reporting but are not scored. The observed churn flag is used only as a validation/diagnostic signal because it would not be known when predicting future churn.

3. Rule-Based Churn-Risk Score

Feature	Condition	Points
Contract	Month-to-month / One Year / Two Year	30 / 10 / 0
Tenure	<6 / 6–12 / 13–24 / >24 months	25 / 15 / 8 / 0
Support tickets	≥5 / 3–4 / 1–2 / 0	25 / 15 / 5 / 0
Subscription	Basic / Pro / Enterprise	10 / 5 / 0
Observed churn	Yes / No (validation only)	10 / 0

Risk bands: Low = 0–29, Medium = 30–59, High = 60–100.

4. Segment Results

Risk Level	Customers	Avg Tenure	Avg Tickets	Monthly Revenue	Churned	Churn Rate
High	7	7.0	4.3	■409.93	7	100.0%
Low	6	32.2	0.8	■759.94	0	0.0%
Medium	2	20.0	1.5	■99.98	0	0.0%

5. Customer-Level Prioritization

Customer	Score	Risk	Contract	Tenure	Tickets	Subscription
CUST-1007	100	High	Month-to-month	3	6	Basic
CUST-1014	100	High	Month-to-month	2	5	Basic
CUST-1003	90	High	Month-to-month	6	5	Basic
CUST-1010	90	High	Month-to-month	4	4	Basic
CUST-1001	80	High	Month-to-month	12	3	Basic
CUST-1005	75	High	Month-to-month	8	4	Pro
CUST-1012	68	High	Month-to-month	14	3	Pro
CUST-1006	33	Medium	One Year	18	2	Basic
CUST-1013	33	Medium	One Year	22	1	Basic
CUST-1002	28	Low	One Year	24	1	Pro
CUST-1009	28	Low	One Year	15	2	Pro
CUST-1008	5	Low	Two Year	48	1	Enterprise
CUST-1015	5	Low	Two Year	40	1	Enterprise
CUST-1004	0	Low	Two Year	36	0	Enterprise
CUST-1011	0	Low	Two Year	30	0	Enterprise

6. Recommendations

- **High risk:** Prioritize immediate retention outreach. Resolve open/support issues, offer an appropriate retention incentive, and encourage a longer contract where suitable.
- **Medium risk:** Use proactive check-ins, monitor support activity, and encourage movement to a longer-term contract or better-fit plan.
- **Low risk:** Maintain engagement and consider loyalty/upsell communication while continuing to monitor changes in tenure, support demand, and contract status.

7. Validation and Edge Cases

The score is additive and bounded by the component rules. Customers with zero support tickets receive zero support points, while customers with long tenure receive zero tenure-risk points. No missing values or duplicate Customer IDs were visible in the supplied 15-row dataset. The model is intentionally transparent so that each score can be traced back to specific business rules. Because the sample is small and synthetic, the score should be treated as a prioritization prototype rather than a statistically validated production prediction model.

8. Conclusion

The segmentation model converts customer attributes into an actionable retention queue. The highest-priority customers are those combining short contracts, shorter tenure, heavier support activity, and lower-tier subscriptions. The accompanying Excel workbook provides the customer-level score, segment, and recommendation along with the complete scoring rules and summary tables.